Subtracting Across a Zero

Answer Key

Grades 2–3   Number and Operations in Base Ten

Quick Answers

1. 155	4. 300	7. 455	10. 513
2. 367	5. 264	8. 430	
3. 178	6. 502	9. 445	

Curriculum

Level: Grades 2–3   Number and Operations in Base Ten
3.NBT.A.2 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*
Difficulty 1–4 of 5 across 10 tagged checks.

Common wrong answers

9. *If they got 555: took the smaller digit from the larger in every column instead of regrouping across the zero*

1. 302 − 147.
Ones: $2-7$ is not possible, and the tens digit is 0, so the tens cannot give a group. **Trade a hundred:** $302 = 2$ hundreds, 10 tens, 2 ones. **Now trade a ten:** 2 hundreds, 9 tens, 12 ones. **Subtract each place:** $12-7=5$ ones; $9-4=5$ tens; $2-1=1$ hundred. $\boxed{155}$
Check by adding: $155+147=302$ ✓

2. 605 − 238.
The ones need a group but there are 0 tens. Trade a hundred: 5 hundreds, 10 tens, 5 ones. Trade a ten: 5 hundreds, 9 tens, 15 ones. $15-8=7$; $9-3=6$; $5-2=3$. $\boxed{367}$
Check by adding: $367+238=605$ ✓

3. 704 − 526.
Trade a hundred: 6 hundreds, 10 tens, 4 ones. Trade a ten: 6 hundreds, 9 tens, 14 ones. $14-6=8$; $9-2=7$; $6-5=1$. $\boxed{178}$
Check by adding: $178+526=704$ ✓

4. Estimate 608 − 293 to the nearest hundred.
608 rounds to 600 (the tens digit 0 is less than 5). 293 rounds to 300 (the tens digit 9 is 5 or more). $600-300=300$. $\boxed{300}$
Note for the grader: the exact difference is 315, so an estimate of 300 is sensible. A student who rounded 293 down to 200 has rounded, not estimated well — send them back to the hundreds digit rule.

5. $500 - 236$.
Both the ones and the tens are 0, so travel left to the hundreds. Trade a hundred: 4 hundreds, 10 tens, 0 ones. Trade a ten: 4 hundreds, 9 tens, 10 ones. $10 - 6 = 4$; $9 - 3 = 6$; $4 - 2 = 2$. $\boxed{264}$
Check by adding: $264 + 236 = 500$ ✓

6. Kim wrote $502 - 138 = 364$. Check by adding.
Addition undoes subtraction, so a correct difference plus the number subtracted must give back the starting number: $364 + 138$. Ones: $4 + 8 = 12$, write 2 carry 1. Tens: $6 + 3 + 1 = 10$, write 0 carry 1. Hundreds: $3 + 1 + 1 = 5$. $\boxed{502}$
Because the check returns 502, which is exactly the number Kim started from, her answer is correct. *Expected sentence:* "The check gives 502, so Kim is right."

7. $800 - 345$.
Trade a hundred: 7 hundreds, 10 tens, 0 ones. Trade a ten: 7 hundreds, 9 tens, 10 ones. $10 - 5 = 5$; $9 - 4 = 5$; $7 - 3 = 4$. $\boxed{455}$
Check by adding: $455 + 345 = 800$ ✓

8. Estimate $702 - 268$ to the nearest ten.
702 rounds to 700 (ones digit 2). 268 rounds to 270 (ones digit 8). $700 - 270 = 430$. $\boxed{430}$
Note for the grader: the exact difference is 434, so this estimate lands within 4 — rounding to the nearest ten gives a much tighter estimate than rounding to the nearest hundred would.

9. Find and fix Rico's mistake in $903 - 458$.
The mistake: in each column Rico subtracted the smaller digit from the larger one ($8 - 3$, $5 - 0$, $9 - 4$) instead of regrouping. Subtraction is not "take the little digit from the big one" — the top number must give the bottom number away, and when the top digit is too small it has to borrow.
Correct work: the ones need a group and the tens digit is 0, so trade a hundred: 8 hundreds, 10 tens, 3 ones. Trade a ten: 8 hundreds, 9 tens, 13 ones. $13 - 8 = 5$; $9 - 5 = 4$; $8 - 4 = 4$. $\boxed{445}$
Check by adding: $445 + 458 = 903$ ✓

10. $1000 - 487$.
Every digit to the left of the ones is 0, so the trade starts at the thousands. $1000 = 9$ hundreds, 9 tens, 10 ones (one thousand traded down through all three places). $10 - 7 = 3$; $9 - 8 = 1$; $9 - 4 = 5$; nothing is left in the thousands. $\boxed{513}$
Check by adding: $513 + 487 = 1000$ ✓